

2(a)(i)	any three from: 1 phagocytosis / endocytosis ; 2 (antigens on) bacteria, attach / bind, to cell (surface) membrane / <u>receptors</u> ; I 'bonding' / binding sites A in context of, opsonins / antibodies, attached to bacteria 3 infolding / invagination / pinching in of membrane / AW ; A pseudopodia / cytoplasm, moves around bacteria 4 fusion of membrane of macrophage (to form vacuole) ; 5 formation of, vacuole / vesicle / phagosome ; 6 active process / uses ATP / uses energy ; R active transport I lysosomes	3
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Question	Answer	Marks
2(a)(ii)	any three from: <i>lysosomes contain enzymes</i> 1 (lysosomes contain enzymes) to, break down / digest / AW, pathogens / bacteria / viruses ; 2 (lysosomes contain enzymes) to break down, worn out / old / non-functioning, organelles / cell components ; A named organelle / worn out cells / old cells / ref to autolysis 3 using, hydrolytic enzymes / hydrolases or any two examples of enzymes ; e.g. nuclease / protease / carbohydrase / lipase / glycosidase / lysozyme 4 isolate harmful enzymes from rest of the cell / membrane-bound to protect rest of cell (from hydrolytic enzymes) ; 5 AVP ; e.g. maintaining acid environment / optimum pH, for enzyme action ref. allows antigen presentation	3
2(b)(i)	any two from: <i>if working and numerical answer shown correctly but no minus sign or decrease stated = 2 marks</i> (number of new cases in 1993) 25 000 – 25 100 and (number of new cases in 2018) 8000 – 10 000 ; percentage change = $\frac{16\,000}{25\,000} \times 100$; answer = - 64(%) / decrease by 64(%) ; A answer between – 60 and – 68 (%)	3

Question	Answer	Marks
2(b)(ii)	<i>any two from</i> 1 (number of new cases overall) decrease ; 2 steep(er) decrease between 1993 and 2001 ; 3 (number of new cases) constant / stable / plateau, between, 2013 / 2014, to 2018 ; 4 (number of new cases) increases in 2015 ; I fluctuation for mp3 and mp4	2
2(c)	<i>any one from:</i> I references to (more) accurate can make, statistical / valid, comparisons between, years / countries ; A takes into account the population sizes of different countries AVP ; e.g. can set targets for the control of TB (e.g. keep below n per 100 000) ref. to supply of, vaccines / drugs / hospital beds / AW can, monitor / evaluate, success of TB control programmes to find out which countries have high, rates / incidence, of TB R 'high numbers'	1

Question	Answer	Marks
2(d)	any two from: 1 <i>Mycobacterium</i> / pathogen / TB bacteria, remains dormant in the body ; A ref. to latent TB / bacterium present in the body but no symptoms 2 treatment takes a, long time / minimum of 6 months ; 3 many people do not complete treatment ; 4 many people do not have access to treatment ; 5 drug-resistant / antibiotic resistant, strains of <i>Mycobacterium</i> / pathogen / TB bacteria ; R 'immune' 6 HIV/AIDS or any other medical condition that increases susceptibility to TB ; e.g. weakened immune system / increased activation of dormant bacteria I TB is opportunistic, infection / disease, without further detail 7 vaccine reluctance ; 8 movement / migration, of people, infected with TB / from countries with high rates of TB ; 9 poor housing / overcrowded living conditions / homelessness ; 10 malnutrition / poor diet ; 11 transmission from cattle by, drinking unpasteurised contaminated milk / eating meat from contaminated cattle ; 12 lack of education / AW, about preventing spread of TB ; 13 AVP ; e.g. vaccination is not effective for adults / AW TB is difficult to diagnose	2